Levers and limits of active control in cross-situational word learning

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Abstract

Learners who choose to hear words they do not yet know can recover much of the cost that a Zipfian vocabulary imposes on learning from random input. Analyses of this benefit, including our own, have idealized the learner's control: it chooses only the target word, on every episode, from the entire vocabulary, at no cost. We relax each idealization in simulations of two cross-situational learners, one that eliminates candidate referents and one that tests a single guess per word. (1) Which lever: with small contexts, choosing the target does most of the work; with 100 referents per context, learning never finishes under Zipfian distractors unless the learner also controls what else is in view. Cross-word mutual exclusivity also rescues learning, but leaves target choice with nothing to do. (2) How much control, and over what: a learner active on a quarter of episodes gains much of the full benefit, and the guess-testing learner can do better with partial control than with full control; but choice sets that over-represent frequent words can make learning more than 10 times slower. (3) When: becoming active late in learning is slightly better than spreading the same autonomy evenly, while stopping early is much worse. (4) At what cost: on real image captions, staying with the current scene beats seeking a new one once switching scenes costs more than an episode or two, by up to 7% for the eliminative learner and 28% for the guess-testing learner. The benefit of active control depends on which levers a learner has and how they combine, not only on how much control the learner has.

Keywords: Word learning; Cross-situational learning; Active learning; Self-directed learning; Zipf's law; Development

Levers and limits of active control in cross-situational word learning

Introduction

Children hear words in a world that is far from uniform. Both the words caregivers say and the objects infants see follow highly skewed, Zipfian distributions (Clerkin, Hart, Rehg, Yu, & Smith, 2017; Lavi-Rotbain & Arnon, 2023), and under such distributions a learner who takes input as it comes must wait a long time for the rarest words. In a companion paper (Hidaka, Torii, & Kachergis, in preparation), we showed that this wait multiplies the number of learning episodes needed by roughly $\ln W$ for a vocabulary of W words, and that a learner who influences its input can recover much of the cost. A learner choosing among scenes can make rare words more likely, and an incremental learner who targets words it does not yet know learns 2–10 times faster than a passive one by the time it knows 50–90% of its vocabulary.

That analysis, like most formal treatments of active learning (Settles, 2012), idealized the learner's control in four ways. The learner chose only which word to hear, not what else was in view. It chose on every episode, from the whole vocabulary, always preferring the least-known word. Its control was the same from the first episode to the last. And obtaining a situation was free. None of these is true of children. Children are active learners (Foushee, Srinivasan, & Xu, 2023; Gureckis & Markant, 2012): they direct their gaze, hands, and requests toward what interests them (Partridge, McGovern, Yung, & Kidd, 2015; Yu, Smith, Shen, Pereira, & Smith, 2009), prefer stimuli of intermediate rather than minimal or maximal complexity (Kidd, Piantadosi, & Aslin, 2012), and adapt their information search to the environment from about age three (Ruggeri, 2022). But their control is partial. Much of what they hear is not directed by them at all. They choose among the objects at hand, not among every object they will ever name. Their capacity for control grows with motor development, which reshapes what infants see and do (Iverson, 2010; Kretch, Franchak, & Adolph, 2014; Walle & Campos, 2014). And, like adults, they do not always choose well (Bazhydai, Wong, Altmann, Jones, & Westermann, 2026; Tauber, Dunlosky, Rawson, Wahlheim, & Jacoby, 2013).

This paper asks how the benefit of active control depends on these features. We organize the question around four aspects of control.

Which lever. Caregivers and children shape not only what is named but what else is in view when it is named: in play, they often arrange for a single object to dominate the child's view (Suanda, Barnhart, Smith, & Yu, 2018; Yu et al., 2009), and children exclude already-known objects as candidate referents for a new word (Grassmann, Schulze, & Tomasello, 2015). Study 1 asks how control over the target word and control over the accompanying distractors combine, and how this depends on the number of referents in view.

How much, and over what. Study 2 asks how much of the benefit survives when the learner is active only some of the time, when it chooses only among a small set of available words, and when it prefers words of intermediate familiarity, as the Goldilocks effect suggests infants do (Kidd et al., 2012).

When. Infants begin with little control over what they experience and gain it as they learn to reach, crawl, and walk. Because active selection is worth more late in learning, when only rare words remain, the timing of control could matter in its own right. Study 3 compares developmental trajectories from passive to active, and the reverse, with constant levels of control.

At what cost. Gaining a new situation takes effort; asking more about the current one may take less. This is the patch-leaving problem of foraging theory (Charnov, 1976), which has been applied to search in memory and other cognitive domains (Hills, Todd, Lazer, Redish, & Couzin, 2015). Study 4 asks, using real image captions, when a learner should stay with the current scene and when it should move on.

Across the four studies, the same theme emerges: active control helps by directing the learner's episodes toward its current bottleneck, and different levers act on different bottlenecks. Levers that act on different bottlenecks need not combine, and constraints that change where the bottleneck lies can change which lever matters. Figure 1 summarizes the four aspects and the main result for each.

General method

All studies use the two incremental learners of the companion paper, an eliminative learner (after Siskind, 1996; Vogt, 2012) and a guess-test learner (after Trueswell, Medina, Hafri, & Gleitman, 2013). There are W words, each paired with one of W referents. In each episode the learner hears one word, the *target*, in a situation containing its referent and $C - 1$ distractor referents. The eliminative learner keeps, for each word, the set of referents present every time that word has been heard, and learns the word once a single candidate remains. The guess-test learner keeps one guess per word, proposing a referent from the current situation that no other word claims, discarding the guess if the word is later heard without it, and learning the word once its guess is correct.

Word and referent frequencies are Zipfian, $p_r \propto r^{-a}$ over frequency rank r, with $a = 1$ unless stated ($a = 0$ is uniform). Two choices define each episode:

- *Target selection.* A *passive* learner's target is drawn by frequency. An *active* learner's target is drawn among words not yet known, in proportion to their frequency. Study 1 also compares a *uniform* active policy, which chooses among the unknown words without regard to frequency; unless stated otherwise, active selection is frequency-weighted.

- *Context selection.* Under *random* context selection, the $C - 1$ distractors are drawn by frequency. Under *familiar* context selection, each referent's frequency is divided by one plus the number of words whose candidate sets still contain it, so that distractors tend to be referents the learner has already ruled out for most words. This formalizes the use of known objects to disambiguate new words (Grassmann et al., 2015), and is consistent with evidence that discourse and attention, not only visual presence, determine which referents are candidates (Frank, Goodman, & Tenenbaum, 2009; Frank, Tenenbaum, & Fernald, 2013). The guess-test learner has no candidate sets, so for it familiar context instead weights each referent's frequency by one plus the number of consecutive times its

word's current guess has been confirmed, favoring referents of words the learner has repeatedly heard.

Unless stated, vocabularies have $W = 1{,}000$ words, and we record the number of episodes until 50%, 90%, and 99% of the vocabulary is learned, with 30 replications per condition. Simulations at $C = 100$ and at $W \geq 10{,}000$ used the companion paper's fast reimplementation of the eliminative learner, which agrees with the reference implementation within sampling error on every condition both can complete. Code and results are available in the project repository (`corpusXSL/analysis`).

Study 1: Which lever? Target selection vs. context selection

The companion paper varies one lever: which word is the target of an episode. But a learner or caregiver also influences what else is in view when a word is heard. This study crosses target selection (passive or active) with context selection (random or familiar), and shows that at larger context sizes the second lever, not the first, decides whether elimination-based learning can finish at all.

We simulated the eliminative learner at $C = 10$ and $C = 100$, on either side of Vogt (2012)'s reported tractability boundary of $C/W \leq 0.1$ (his C/M), for uniform and Zipfian vocabularies of $W = 1{,}000$ words, and recorded episodes to learn 99% of the vocabulary. Runs were capped at 5×10^8 episodes, and capped runs are reported as such.

Results: context size decides which lever matters

Table 1 and Figure 2 summarize the results at $a = 1$. At $C = 10$ ($C/W = 0.01$), active target selection does most of the work (19.1x), and familiar-context selection adds only a modest further benefit (2.0x alone, 28.6x combined), because elimination is already fairly effective when the distractor set is small.

At $C = 100$ ($C/W = 0.1$), random-context learning does not merely slow down under a Zipfian distribution: it fails. No passive or active run with random distractors reached even half the vocabulary within 5×10^8 episodes. The reason is the eliminative

learner's requirement that every competing referent be absent from at least one of a word's contexts. With 99 frequency-weighted distractors per context, the most frequent referent ($p \approx 0.13$ at $a = 1$) is absent from any given context with probability only about 5×10^{-10} (computed exactly[1]), so every other word would need on the order of 2×10^9 exposures merely to rule out that one competitor. Active target selection cannot help, since it changes which word is the target but not which distractors accompany it. Familiar-context selection, which does change the distractors, restores learning (446,962 episodes), and once distractors are controlled, active target selection adds a further 28-fold advantage (15,819 episodes). At $C = 100$, controlling what else is in view is a precondition for learning at all.

This failure is a matter of degree of skew, not an all-or-none property of $C = 100$. Sweeping the Zipf exponent at $C = 100$ (Figure 3, right) shows passive random-context learning slowing steeply but continuously, from 9,904 episodes at $a = 0$ to 40,456 at $a = 0.5$, 1.45 million at $a = 0.7$, 47 million at $a = 0.8$, and beyond our 10^8-episode cap from $a = 0.85$ on, tracking the exact probability that the most frequent referent is absent from a given context (0.17 at $a = 0.5$, 2.6×10^{-4} at $a = 0.8$, 1.5×10^{-6} at $a = 0.9$). Active target selection helps throughout this range (e.g., 12x at $a = 0.8$) but succumbs in turn (38 million episodes at $a = 0.85$, beyond the cap at $a = 0.9$). Only familiar-context selection keeps learning fast as skew grows (at $a = 0.9$: 106,039 passive and 10,646 active episodes).

Context size, not the ratio C/W

The same failure appears at larger vocabularies (Table 2). $C = 100$ at $W = 10{,}000$ has the same ratio, $C/W = 0.01$, as $C = 10$ at $W = 1{,}000$, where learning is easy, and $C = 100$ at $W = 60{,}000$ has a smaller ratio still (0.0017). Yet at both sizes, with

[1] By the exponential-clock representation of weighted sampling without replacement: each referent j receives an independent clock $E_j \sim \text{Exp}(p_j)$, the context's distractors are the $C - 1$ referents whose clocks ring first, and referent r is absent iff at least $C - 1$ other clocks ring before E_r. The resulting one-dimensional integral over a Poisson-binomial tail agrees with direct simulation wherever the probability is large enough to simulate; see `corpusXSL/analysis/c100_absence_probability.R`.

random distractors, no run, passive or active, learned even 10% of the vocabulary within 10^8 episodes (10 replications per condition). With 99 distractors per context, the most frequent referent ($p \approx 0.10$ at $W = 10{,}000$ and 0.086 at $W = 60{,}000$) is absent from a given context with probability only about 1.5×10^{-6} and 2.7×10^{-5} respectively, so the rarest words would need on the order of 10^{10} episodes or more merely to rule out that one competitor. Familiar-context selection again restores learning. Under Zipfian distractors, then, whether elimination is feasible depends on the absolute context size together with the skew of the referent distribution, rather than on the ratio C/W alone.

C/W does, however, shape how large the active advantage is once learning is possible. With familiar context at $C = 100$, the active advantage is 28-fold at $W = 1{,}000$ but only 7.6-fold at 10,000 and 9.3-fold at 60,000. Recording how many exposures each word needed before it was learned (10 replications per condition) shows why. Target selection does not change how many exposures a given word needs: passive and active learners need the same number, frequency band by frequency band. What differs is how each learner pays for them. An active learner never spends an episode on a word it already knows, so its total is essentially the sum of all words' needs, and under familiar context at $C = 100$ that sum is dominated by the most frequent words, whose competitors are other frequent referents: the top 1% of words need a median of 109–229 exposures each, the bottom half only 5–7. A passive learner instead waits for its last few rare words to be sampled often enough, so each exposure a rare word needs costs it roughly the inverse of that word's sampling probability. At $W = 1{,}000$ these last words are unusually hard–the last 1% learned need a median of 34 exposures, against 5–6 at $W = 10{,}000$ and 60,000–which inflates the passive learner's time, and hence the speedup. Varying C at fixed M indicates that this hard tail grows with the share of all referents present in each context (5 replications per condition): at $W = 1{,}000$, the last 1% of words need a median of 3, 6, and 34 exposures at $C/W = 0.01$, 0.03, and 0.1, and at $W = 10{,}000$, 6, 10, and 297 at the same ratios. When each context contains a tenth of all referents, a competitor that survives one exposure has a good chance of reappearing in the next.

Which unknown word to target

An active learner must also decide which of the words it does not know to target. Our default draws them in proportion to frequency; a uniform policy ignores frequency. Table 3 compares the two for the eliminative learner, using the fast reimplementation (30 replications per cell, none censored).

With random context the policy makes no difference: the number of exposures a word needs does not depend on which other words are already known, so any learner that targets only unknown words takes about the total number of exposures needed, whatever their order. With familiar context the policy matters, slightly at $C = 10$ (8%) and greatly at $C = 100$, where uniform targeting halves the learning time and roughly doubles the active advantage, from 30-fold to 58-fold. The reason is the dilution of the familiarity signal by words not yet heard, analyzed in Study 2: after 5,000 episodes, frequency-weighted targeting had left 239 words unheard on average, uniform targeting about 6 (10 replications each). With that dilution removed, by computing familiarity from heard words only, frequency-weighted targeting reached 99% in 7,698 episodes, close to uniform targeting (8,041). The active advantages reported for familiar context elsewhere in this paper use frequency-weighted targeting and are therefore conservative. For the guess-test learner, the policy matters even with random context (Study 2): uniform targeting is about twice as fast at $C = 10$ and four times as fast at $C = 100$.

A learner that does not need full elimination

The guess-test learner of the companion paper, which keeps a single guess per word and counts a word as learned once its guess lands on the true referent, never needs every competing referent to be absent from some context. Accordingly it still learns at $C = 100$, $a = 1$ with random distractors (326,494 episodes passive, 176,578 active), where the eliminative learner never finishes, and active target and familiar-context selection give it only modest speedups (1.8x and 3.8x). Combining active target selection with familiar-context selection, however, often prevented this learner from finishing: in Study 2, 22 of 30 such runs did not learn 99% of the vocabulary within

5×10^6 episodes, while every passive run finished (85,818 episodes on average). We return to this failure in Study 2.

Cross-word mutual exclusivity

Every learner above narrows a word's candidates independently. Reisenauer, Smith, Smith, and Blythe (2013) show that a cross-word mutual exclusivity constraint–confirming one word's meaning immediately excludes that referent for every other still-unlearned word–can rescue an elimination-based learner from exactly the slowdown seen at large C. We added this rule to the eliminative learner, with cascades of further learning events allowed within an episode, and compared it against the independent model at $W = 1{,}000$, $a = 1$, random context (Table 4; 30 replications per cell at $C = 10$, and 200 at $C = 100$, where learning time is heavy-tailed; no replication was censored).

At $C = 100$, mutual exclusivity delivers the predicted rescue: both passive and active learners now reach 99% of the vocabulary, in well under a million episodes on average. The trajectory is unusual, however. Essentially the whole vocabulary is learned in a single cascade: in 396 of 400 replications, the 50%, 90%, and 99% checkpoints were all reached within 10 episodes of one another, after a long initial period in which almost nothing was learned. And active target selection confers little benefit: 686,349 versus 771,125 episodes (medians 441,212 and 572,126; $z \approx 1.3$).

Both features have the same explanation. Nothing can cascade until at least one word has been learned by ordinary elimination, which requires ruling out every competitor for that word–least severe for the most frequent word, whose own referent is the competitor otherwise hardest to rule out. Once that first word is learned, its referent is removed from every other word's candidate set, which unblocks the next, and so on. The wait for the first learning event dominates the total time, and during that wait active target selection is inert: its advantage comes from never spending an episode on an already-known word, and while no word is yet known, that restriction does nothing. Holding the random seed fixed, active and passive target selection produce

identical target sequences until the first word becomes known. Mutual exclusivity and active target selection, both beneficial elsewhere, therefore do not compound at large C–a second case, alongside guess-test's bimodality, of levers that fail to compose.

Study 2: How much autonomy, and over what?

In the companion paper and in Study 1, an active learner is idealized in two ways: it exercises choice on every episode, and whenever it does, it can choose among the entire vocabulary, preferring any word it does not yet know. Real learners exercise control only some of the time; they choose among whatever happens to be available (the toys in the room, not every object they will ever name); and they may not simply prefer the least-known item, since infants preferentially attend to stimuli of intermediate familiarity (the "Goldilocks effect"; Kidd et al., 2012). This study relaxes each idealization in turn.

Setup

We extended the eliminative learner's target selection in three ways. *Partial autonomy*: each episode is active with probability active_prob $\in \{0.25, 0.5, 0.75\}$ and passive otherwise, a constant blend of active and passive selection. *Goldilocks targeting*: among not-yet-known words, the learner prefers those it has encountered about twice, weighting each by $\exp(-(n-2)^2/8)$ times its frequency, where n is the number of times the word has been the target so far. *Bounded choice*: on each active episode the learner may choose only among a choice set of $k \in \{100, 50, 20, 5\}$ words, drawn without replacement in proportion to frequency (so that commonly-encountered things are more likely to be at hand), preferring an unknown word within the set and making a passive draw from the set if it contains none.

We ran each condition, together with pure passive and fully active selection, in four settings for a Zipfian vocabulary of $W = 1{,}000$ words: $C = 10$ with random and with familiar context selection ($a = 1$), and two settings at $C = 100$. Study 1 shows that at $C = 100$, $a = 1$, the eliminative learner never completes with randomly-composed contexts, even under fully active selection, so there we use the two $C = 100$ settings in

which learning does complete: familiar context at $a = 1$, and random context at $a = 0.75$, the steepest skew at which random-context learning finished in every run. All conditions were simulated with the companion paper's fast reimplementation of the eliminative learner, which we validated for each of these target-selection rules against the reference implementation (200 replications per condition at $C = 10$, 30 at $C = 100$).

Results

Table 5 and Figure 4 report each condition's episodes to learn 99% of the vocabulary relative to fully active selection in the same setting.

Partial autonomy shows steeply diminishing returns. In every setting, a learner who is active on only a quarter of episodes already captures most of active selection's benefit: measured on a log scale, between two-thirds and four-fifths of the full passive-to-active speedup (e.g., at $C = 10$ with random context, 7.1x of a possible 19.0x). Each further increment of autonomy buys less. Study 3 asks whether, beyond its amount, the timing of autonomy matters.

Goldilocks targeting matters in only one setting, and not because of intermediate familiarity. Preferring words encountered about twice makes no difference in three of the four settings (0.996–1.04x). At $C = 100$ with familiar context it nearly halves learning time (0.56x), but the mechanism test below shows that this gain mostly disappears (0.93x) once familiarity is computed from encountered words only. Its benefit there comes from spreading early exposures across more of the vocabulary, which speeds up familiar-context selection, not from targeting words of intermediate familiarity as such.

The cost of bounded choice depends on the setting. In three settings, restricting choice costs little until the choice set is small, and costs more the smaller it is (1.03–1.09x with 100 words to choose from, rising to 2.8–4.1x with 5). At $C = 100$ with familiar context, the pattern is different: even a choice set of 100 makes learning 5.5 times slower, and the cost is not monotone in the size of the set (11.3x with 20 words, 10.4x with 5).

A diagnostic isolates the cause of this last pattern. Drawing the choice set uniformly rather than in proportion to frequency, and otherwise leaving everything unchanged, removes the cost almost entirely at $C = 100$ with familiar context: 15,124 episodes with a uniform set of 100 words and 15,894 with 20, against 15,876 for unrestricted choice (30 replications each). The cost therefore comes not from restricting choice as such but from choice sets that over-represent frequent words: when the set is drawn by frequency and the target is then chosen from it by frequency again, a rare word's chance of being targeted scales roughly with the square of its frequency, and rare words are rarely targeted at all. Together with the Goldilocks result, this suggests that at $C = 100$ with familiar context, learning is fastest when exposures are spread across the whole vocabulary, including its rare words.

Mechanism: unseen words dilute the familiarity signal

Why should spreading exposures matter, and only with familiar context at large C? Familiar-context selection weights each candidate distractor r by $1/(n_r + 1)$, where n_r is the number of words that still hold r as a possible meaning. A word that has never been the target of an episode has eliminated nothing, so it still holds every referent as a possible meaning, and each such unseen word adds the same amount to every referent's n_r. While many words are unseen, this shared count swamps the differences between referents, and familiar-context selection behaves almost like random selection. On this account, familiar context only starts to help once most words have been encountered at least once, and policies that concentrate targets on frequent words delay that point.

We tested this account in two ways, both at $C = 100$, $a = 1$ with familiar context (Table 6, Figure 5). First, we removed the dilution by computing each referent's familiarity from encountered words only, i.e. discounting the contribution of unseen words. If dilution is the mechanism, this should largely remove the differences between policies, and it does. The frequency-weighted choice set of 100 words goes from 5.7 times slower than unrestricted choice to 1.3 times; Goldilocks targeting goes from 0.57 to 0.93. At the 50% checkpoint, the choice-set cost disappears entirely (5,547 episodes

vs. 5,127 unrestricted, down from 84,671). What remains for small frequency-weighted choice sets (3.3x for 20 words) arises late in learning, and reflects the separate, direct problem that rare words are seldom in the choice set and so are seldom targeted. Removing the dilution also halves learning time for unrestricted active selection (15,770 to 7,803 episodes), while leaving passive sampling essentially unchanged (466,339 to 451,567), whose bottleneck is instead how rarely its rare words are targeted.

Second, we tracked each policy's trajectory under the original weighting. The frequency-weighted choice set leaves many words unencountered for a long time–634 after 10,000 episodes, against 63 for unrestricted choice and 34 for a uniform choice set–and throughout this period, 96–98% of a target's remaining competing referents reappear among its distractors, escaping elimination, against 66–84% for the other policies. Goldilocks targeting, which favors words encountered only once or twice, brings the number of unencountered words down fastest (33 after 5,000 episodes) and has the lowest competitor survival (about 50%).

Why only at large C? At $C = 10$ elimination is fast even with random distractors, so familiar-context selection adds relatively little (Study 1 finds a 2-fold benefit at $C = 10$, against a difference between learning and not learning at all at $C = 100$), and a policy that delays its benefit loses correspondingly little.

These tests also suggest a modeling point. The dilution arises from treating a never-encountered word as "still considering every referent", which is the natural bookkeeping for an eliminative learner, but not the only plausible one: a learner that simply ignores words it has not yet heard when judging how familiar an object is learns twice as fast under unrestricted active selection, and is far less sensitive to how its choice set is composed.

The guess-test learner

We repeated the study with the guess-test learner, at $a = 1$ in all four settings, since this learner finishes at $C = 100$ even with random context (Study 1). Runs were capped at 5×10^6 episodes. Table 7 gives mean episodes to learn 99% of the vocabulary.

Partial control can beat full control. With random context, partial autonomy again shows diminishing returns, though less steeply than for the eliminative learner: being active on a quarter of episodes gains 59% (at $C = 10$) and 42% (at $C = 100$) of the full speedup on a log scale, against two-thirds to four-fifths for the eliminative learner. With familiar context, partial autonomy is as good as or better than full autonomy. At $C = 10$, learners active on half or three-quarters of episodes are 18–24% faster than fully active ones. At $C = 100$, full active selection fails: 22 of 30 runs did not finish, whereas every learner active on 25–75% of episodes finished, in 68,658–80,128 episodes, faster than passive learning. The companion paper found a related limitation of purely active selection: a learner that never returns to words it believes it knows cannot correct confident errors. The diagnostics below trace this failure to a feedback loop between wrong guesses and familiar-context selection, which any episodes spent on known words break.

Goldilocks targeting helps the guess-test learner. Preferring words heard about twice more than doubles the speed of guess-test learning in three of the four settings: 3,632 vs. 7,814 episodes at $C = 10$ with random context, 3,594 vs. 7,787 with familiar context, and 63,772 vs. 179,983 at $C = 100$ with random context. It fails, like full active selection, at $C = 100$ with familiar context (29 of 30 runs censored). The diagnostics below show that the benefit comes from spreading targets more evenly over the unknown words, not from intermediate familiarity as such.

Bounded choice is costly and erratic. At $C = 10$, frequency-weighted choice sets produce heavy-tailed learning times: median times are similar to or shorter than with unrestricted choice (e.g., 5,848 episodes with 20 words vs. 7,600 unrestricted, random context), but occasional slow runs inflate the means (with 50 words, mean 13,360, median 5,952). At $C = 100$ with random context, choice sets of 20–100 words make learning 8.8–15.5 times slower than unrestricted choice, and a set of 5 words only 2.8 times slower–the same non-monotone pattern as for the eliminative learner with familiar context. Here there is no familiar-context selection, so the dilution mechanism of the eliminative learner cannot be the explanation; the diagnostics below identify a different

one.

Mechanism: blocked words and self-confirming errors

To find out why these policies help or fail the guess-test learner, we reimplemented it with additional bookkeeping (`corpusXSL/analysis/diagnose_guesstest.R`) and ran a small number of instrumented replications, with and without targeted interventions.

Blocked words. For the guess-test learner, a word can be learned only when it is the target and its own referent is not held, as a wrong guess, by another word. At $C = 100$ this is the norm rather than the exception: in instrumented runs with random context, 97.5–99.9% of active episodes targeted an unknown word whose referent was held by another word. Such a word is learned only after the word holding its referent is itself targeted, in a context without that referent, and drops its guess. Learning speed therefore depends on how often the words holding others' referents are targeted.

Familiar context at $C = 100$: errors that confirm themselves. For the guess-test learner, familiar-context selection weights each referent by how often its word's current guess has been confirmed, and the learner cannot tell whether that guess is right. In four of six instrumented fully active runs that had not finished after 10^6 episodes, every one of the 12–17 remaining unknown words held a wrong guess on another unknown word's referent, so each word's own referent was held by another unknown word. A fully active learner never targets known words, so their confirmation counts stay at zero, while each surviving wrong guess raises the weight of its word's referent. By the end, more than 99.99% of the familiarity weight lay on referents of unknown words. Contexts were then filled with exactly the referents these wrong guesses claimed, so the guesses were confirmed rather than disconfirmed, and the set of mutually wrong guesses sustained itself. Two interventions confirm this account. Spending a quarter of episodes rehearsing a known word, chosen by frequency, let all six runs finish, in 48,706–59,042 episodes–faster than passive (85,818) or partially active learning (68,658–80,128). Computing familiarity only from confirmations of known words, an oracle the learner does not have, also let all six runs finish (148,335–224,512 episodes). Passive episodes

help because they mostly rehearse known, frequent words, raising the familiarity of their referents and diluting the self-confirming set. Goldilocks targeting fails in this setting because it, too, never targets known words.

Choice sets at $C = 100$: frequency counted twice. When the choice set is drawn in proportion to frequency, and the target is then chosen within it in proportion to frequency again, a word's chance of being targeted grows roughly with the square of its frequency. Estimating the effective exponent b in $P(\text{target}) \propto p^b$ by simulating the target-selection rule directly gives $b \approx 1$ for unrestricted choice and $b = 1.6$–1.9 for choice sets of 20–100 words while many words are unknown. Small sets rarely contain more than one unknown word once learning is under way, so for them b falls back toward 1 (1.06 for a set of 5 with 100 words unknown, 1.25 for a set of 20), at the cost of episodes spent on known words when the set contains no unknown word. This explains why intermediate sets cost the most. Steep weighting is costly for the guess-test learner because of blocking: it keeps choosing frequent blocked words and seldom chooses the rarer words that block them. At the end of instrumented runs with sets of 50 or 100 words, blocked words had typically been targeted tens of thousands of times, and the words blocking them far less often, usually only tens of times. Keeping the frequency-weighted set but choosing uniformly within it removes the cost: with sets of 100 words, learning took 75,844–85,409 episodes instead of 1.6–1.7 million.

Spreading targets helps. The same account explains the benefit of Goldilocks targeting, which avoids returning to words that have been targeted many times without success: at $C = 10$, only 2–5% of its active episodes targeted a word already targeted ten or more times, against 47–61% for unrestricted active selection. Spreading targets uniformly over the unknown words helps even more: it learned 99% of the vocabulary in 3,833 episodes at $C = 10$ and 43,904 at $C = 100$, against 7,814 and 179,983 for frequency-weighted active selection (six replications each). For this learner, then, frequency-weighted targeting–the default active policy throughout our work–is far from optimal.

Scope

Two conclusions seem robust across the settings and both learners we tested: partial autonomy has diminishing returns, and much of active selection's benefit survives a learner who exercises it only occasionally–for the guess-test learner, partial autonomy can even outperform full autonomy. The results for Goldilocks targeting and bounded choice are more conditional. Their size, and at $C = 100$ with familiar context their direction relative to other settings, depends on context size and on whether distractors are controlled, and at least for bounded choice, on what the available choice set contains rather than merely how large it is. If children's accessible objects are, like the objects they encounter generally, dominated by common ones, the composition of what is at hand may matter as much as the freedom to choose among it. We have tested two learning mechanisms, four settings, and frequency-weighted or uniform choice sets (the latter for the eliminative learner only); how real choice sets are composed, and how large they are, are empirical questions we do not address.

Study 3: A developmental passive-to-active trajectory

Studies 1 and 2 hold the degree of autonomy fixed for the whole simulated lifetime of the learner. Real infants do not work this way: a newborn is carried from room to room with essentially no say over which scenes it experiences, and only gradually acquires the visual, postural, and locomotor control–head-turning, pointing, reaching, eventually crawling and walking–that lets it actively select what to attend to. This study asks whether that developmental trajectory itself, quite apart from its eventual endpoint, has consequences for how much of active selection's benefit a learner realizes.

Motivation and setup

The companion paper's completion-checkpoint analysis already implies an answer is possible: the active/passive gap widens over the course of learning (roughly 4-fold at 50% completion, 9-fold at 90%, 19-fold at 99%, for the eliminative learner at $C = 10$), meaning active selection's marginal value is low early–common words get learned

quickly under either policy–and high late, when only rare stragglers remain. If a fixed "active budget" is worth more late than early, a learner who starts passive and becomes active over time should outperform a constant blend that spends the same budget evenly, and should do considerably better than a learner who front-loads the same budget into the early, low-value period.

We extended the eliminative learner with an optional `active_prob_fn`: a function of the episode count, re-evaluated every episode, in place of a constant `active_prob`. We compare a forward, linear ramp from passive to active, active_prob$(t) = \min(1, t/T)$, against its mirror image, a reverse ramp from active to passive, active_prob$(t) = \max(0, 1 - t/T)$. Over their first T episodes the two ramps are active equally often, but the reverse ramp places that activity where the checkpoint analysis says it is worth least. Both are indexed by episode count rather than fraction of vocabulary learned, since real motor and attentional development runs on its own maturational clock and is not contingent on how much vocabulary a particular child happens to have acquired so far; this makes the ramp's timescale T a free parameter, swept here (at $W = 1{,}000$, $C = 10$, $a = 1$, 30 replications per condition) rather than fixed at a single guessed value.

Results

Table 8 gives the reference points–pure passive and active selection, and constant blends at 13 intermediate levels of autonomy–and Figure 6 shows both ramp directions across seven timescales, $T \in \{1{,}000, 2{,}000, 3{,}000, 5{,}000, 8{,}000, 12{,}000, 20{,}000\}$ episodes.

Forward ramps are slower the longer the learner stays passive, rising smoothly from 4,875 episodes at $T = 1{,}000$, within 5% of pure active selection, to 10,573 at $T = 20{,}000$. Reverse ramps behave very differently. At $T = 1{,}000$ the reverse ramp takes 87,550 episodes, close to pure passive selection (89,857), and it stays near the passive level through $T = 8{,}000$ (57,248) before falling sharply by $T = 12{,}000$ (6,049). The threshold lies where the ramp's midpoint, $T/2$, is close to the time a fully active learner needs to finish (4,681 episodes). When T is large, a learner who starts active

finishes before its autonomy has decayed far. When T is small, autonomy decays long before the vocabulary is complete, and the long remainder of the run is passive.

The contrast between directions at a given T confounds timing with the total amount of autonomy, because a learner who becomes active stays active for the rest of its run, while one who becomes passive stays passive. At $T = 1{,}000$, the forward learner is active on about 90% of its episodes and the reverse learner on fewer than 1%. To isolate timing, we compared each ramp with a constant blend of the same autonomy. For each run we computed its realized autonomy–the mean proportion of active episodes over the episodes it actually ran, which is exact because the ramp depends only on the episode count–and compared each ramp's mean episodes with the constant-blend curve of Table 8 at the ramp's mean realized autonomy, interpolating on a log scale. Forward ramps were 4–13% faster than constant blends of matched autonomy, and the gain grew with the timescale: 4% at $T = 1{,}000$ (autonomy 0.90), 11% at $T = 8{,}000$ (0.46), and 13% at $T = 20{,}000$ (0.26). Reverse ramps that lose control before the vocabulary is learned ($T \leq 8{,}000$) were 1.2 to 2.3 times slower than matched blends, the largest cost at $T = 5{,}000$–$8{,}000$ (autonomy 0.03–0.07); those that finish while still mostly active ($T \geq 12{,}000$) were within 4% of them. Timing therefore matters in the direction the checkpoint analysis predicts–autonomy is worth more late in learning than early–but for a learner who becomes active, its effect is modest compared with the effect of how much of the run is spent under control.

The guess-test learner

The guess-test learner shows the same pattern. With the same ramps and a matching set of constant blends (30 replications each; passive 37,944 episodes, fully active 8,386), forward ramps were 3–15% faster than constant blends of matched autonomy, the gain again largest at long timescales, and reverse ramps were 1.2 to 1.8 times slower, most at $T = 8{,}000$. The threshold for reverse ramps lies at a longer timescale than for the eliminative learner, between $T = 12{,}000$ (19,636 episodes) and 20,000 (11,514), consistent with the $T/2$ rule given this learner's longer active

completion time. Partial autonomy again shows diminishing returns: a constant blend with 25% active episodes captures 62% of the full speedup on a log scale.

Scope

These results support a narrow claim. A developmental trajectory from little control to more control loses little of active selection's benefit if control arrives quickly relative to the time needed to learn the vocabulary, and it is slightly more efficient than the same amount of control spread evenly. Losing control early is much more costly. We make no claim about the timescale of real infant development, and we have tested only linear ramps, indexed by episodes rather than vocabulary or age, for two learning mechanisms at one context size and Zipf exponent. Estimating how quickly children's self-directed exploration increases, for example from egocentric video recorded across the transitions to crawling and walking (Kretch et al., 2014; Sullivan, Mei, Perfors, Wojcik, & Frank, 2021), is a natural next step.

Study 4: Staying or moving on: scene-switching cost on real image captions

The companion paper tests target selection on real, naturalistic co-occurrence structure using Flickr8k (Hodosh, Young, & Hockenmaier, 2013): 8,108 photographs, each independently described by five human-written captions, but its learner always moves on to a new photo after each caption. Real captions let us ask a further question the synthetic simulations cannot pose at all: beyond choosing *which* word to target, a learner who can request another caption of the scene already in view, rather than only ever moving on to a new one, faces a second active choice with no synthetic analogue–and it is not obvious in advance which side of that choice should win.

Setup

We treat each caption as a situation: a target word plus whichever other content words co-occur in that sentence, playing the role of distractors. After excluding function words (articles, prepositions, pronouns, and similar) via a hand-curated stoplist and words occurring only once, 4,318 content words remain, averaging 5.76 per caption–a

naturally-occurring analogue of context size C, close to the companion paper's own $C = 10$ condition. A word is eliminative-learnable if the intersection of the word-sets of every caption in which it appears narrows to the word itself; 83.3% (3,598 words) are learnable this way, and the remainder are stuck with at most 8 residual confusable alternatives, not a pathological blowup.

We compare three policies. *Passive*: sample a photo uniformly at random, reveal one of its captions at random, and always move on. *Active, move-on*: target the rarest not-yet-learned learnable word, sample a photo containing it, reveal one caption, and always move on. *Active, linger*: as move-on, but remain on the current photo–revealing another of its remaining captions–for as long as doing so is still shrinking some unknown word's candidate set, moving on only once the current photo stops paying off. A `move_cost` parameter charges additional episodes for switching to a new photo, formalizing the idea that eliciting another description of the scene already in view may be cheaper than obtaining a new scene built around a specific word; `move_cost`= 0 makes the two active policies differ only in information, not economy.

Results

Table 9 and Figure 7 report mean episodes to reach 50%, 90%, and 99% of the learnable vocabulary (30 replications per cell). The companion paper reports the passive vs. active move-on comparison at zero switching cost as its real-data test of target selection (23-fold at 99%); here we focus on the cost of switching. Even at the highest scene-switching cost we tested, both active policies still finish 99% of the vocabulary in under half the episodes passive sampling requires.

The linger-vs-move-on choice itself shows a genuine crossover rather than one policy dominating throughout. At `move_cost` 0 and 1, always moving to a fresh photo wins (by 13% and 3% at 99%): a new scene supplies more diverse elimination context than a second, independently-written description of the same one, so the extra information from a new scene outweighs the (zero or small) cost of obtaining it. Between `move_cost` 1 and 2, the ranking flips, and lingering's advantage widens as

switching gets more expensive: lingering is 1% faster than moving on at `move_cost`=2, 3% at 3, 5% at 5, and 7% at 10. In other words, whether "ask for more about what's already in view" is the better strategy is not fixed; it depends on the relative cost of obtaining a new situation versus refining the current one, exactly the kind of parameter the companion paper's synthetic simulations, which never separated situation-content from situation-acquisition cost, could not have exposed.

This result should be read as a demonstration that the mechanism generalizes to real co-occurrence structure and that the linger-versus-move-on tradeoff is a real, locatable phenomenon, not as a claim about what value `move_cost` actually takes for a child or caregiver in practice–we have no direct estimate of that value, and supplying one is a natural target for future work. It is also worth being explicit that Flickr8k captions are adult-crowdsourced descriptions of generic photographs, not child-directed speech or a child's own visual experience, so this study speaks to whether the mechanisms in the companion paper behave sensibly on naturalistic language statistics, not to developmental realism specifically.

The guess-test learner

We also ran the guess-test learner on the captions. Each caption updates every content word it contains, in random order: a word whose guess is absent from the caption discards it, and a word without a guess proposes a word in the caption that no other word claims. Learning was scored on the same 3,598 words. The eliminative learner's active policy, which always targets the rarest word not yet learned, deadlocks this learner: a rare word whose wrong guess appears in every caption containing it can never be corrected, and a rarest-first learner keeps returning to it. The active guess-test learner therefore draws its target among unlearned words in proportion to their frequency, as the synthetic learners do, and keeps that target while it lingers on a photo. Runs were capped at 10^6 episodes (Table 10).

Three results differ from the eliminative learner. First, active selection does not complete the vocabulary: active runs level off at 97.5–98.9% of the learnable words,

while every passive run reaches 99%. Captions are fixed, so a word that has guessed a word present in all of its captions is never corrected; passive sampling, which hears frequent words most often early on, presumably lets those words claim their own referents first and leaves rare words fewer wrong options. Second, at zero switching cost active selection helps much less than for the eliminative learner, 1.2, 2.3, and 2.7 times faster than passive learning at 50%, 90%, and 95% of the vocabulary, and once switching costs two or three episodes, active learning reaches 95% of the vocabulary later than passive learning does. Third, lingering matters much more. The policies are within 7% of each other at zero cost, but from `move_cost`=1 on, lingering reaches 95% of the vocabulary in 16–28% fewer episodes than moving on, and also ends with more words learned. The crossover thus lies between `move_cost` 0 and 1 for this learner, rather than between 1 and 2.

General discussion

The companion paper established that active control over learning situations can recover much of the cost of a Zipfian vocabulary. The four studies here ask what happens to that benefit when control is closer to what children have. The answers depend less on how much control a learner has than on which lever it acts on and where the learner's bottleneck lies.

Levers act on bottlenecks

An eliminative learner can be held up in two ways. It can wait for rare words to be heard often enough, or it can wait for frequent competitors to be absent from a word's contexts. Target selection removes the first bottleneck, since an active learner never spends an episode on a word it already knows. Context selection removes the second, since it fills contexts with referents already ruled out for most words. Which bottleneck dominates depends on the number of referents in view and on the skew of their distribution (Study 1). With 10 referents in view, competitors are easily ruled out, and target selection does most of the work. With 100, the most frequent referents are almost never absent from a context, and learning never finishes unless distractors are

controlled–whether by familiar-context selection or by cross-word mutual exclusivity. Once distractors are controlled, target selection again adds a large benefit.

This view also explains when levers fail to combine. Mutual exclusivity removes the competitor bottleneck by a cascade that begins with the first word learned; before that word is learned, target selection has nothing to act on, because no word is yet known, and it adds almost nothing (Study 1). Frequency-weighted choice sets restore the rare-word bottleneck that target selection removed, and they also delay familiar-context selection, because familiarity is computed over words, and words not yet heard dilute the signal (Study 2). For the guess-test learner, combining active target selection with familiar-context selection at $C = 100$ usually prevented learning from finishing, although each helps on its own: because the learner never revisits known words, the familiarity signal comes to reflect only its own wrong guesses, which then confirm themselves (Study 2). In each case, two forms of control that help separately interfere because one changes the state that the other depends on.

Partial, constrained, and developing control

The most robust result is that partial control goes a long way. An eliminative learner active on a quarter of episodes gains between two-thirds and four-fifths of the full benefit, on a log scale, in every setting tested, and a guess-test learner 42–96% (Study 2). For the guess-test learner, partial control can be better than full control: with familiar context, learners active on half or three-quarters of episodes beat fully active ones, and at $C = 100$ every partially active learner finished while most fully active ones did not. On real captions, too, a fully active guess-test learner stopped short of the whole vocabulary while a passive one did not (Study 4). In the familiar-context case we traced the benefit of passive input to rehearsal: episodes spent on known words keep the familiarity signal anchored in what the learner has actually learned, and rehearsing known words on a quarter of episodes was enough to prevent the failure. Together with the companion paper's finding that a purely active learner cannot correct confident errors, this suggests that some input outside the learner's own choices may do

more than dilute active control: it can protect a learner from errors that full control entrenches. This is encouraging for the developmental relevance of active learning, because children's control over their input is intermittent: much of what they hear is overheard, initiated by others, or about things they did not choose.

What a child chooses among may matter more than how often they choose. Restricting choice to small sets of words costs little when the sets are drawn uniformly from the vocabulary, but a great deal when they over-represent frequent words, as the objects within a child's reach plausibly do (Clerkin et al., 2017). Preferring words of intermediate familiarity (Kidd et al., 2012) depends on the learner. For the eliminative learner it made no difference in most settings, and where it helped, the gain came from spreading early exposures across the vocabulary. For the guess-test learner it more than doubled the speed of learning in three of four settings, because it spreads targets over the unknown words; spreading them uniformly was faster still (Study 2). How a learner distributes its choices among the words it does not know can matter as much as whether it chooses them: uniform targeting also doubled the eliminative learner's active advantage with familiar context at $C = 100$ (Study 1), although it made no difference with random context.

Timing matters, in the direction expected from the growing value of active selection over learning, but modestly (Study 3). A learner who becomes active over time does 3–15% better than one with the same autonomy spread evenly, and a learner who starts active and loses control before the vocabulary is learned does up to 1.8 (guess-test) to 2.3 (eliminative) times worse. The largest effect of a developmental trajectory is through how much of the learner's run is spent under control, which depends on how quickly control arrives relative to how quickly the vocabulary is learned.

The cost of new situations

On real captions, whether to stay with the current scene or seek a new one depends on the relative cost of the two (Study 4): when switching scenes is free or cheap, a new scene is worth more than another description of the same one, and once it

costs two or more episodes, staying is better. For the eliminative learner the difference was small, at most 7% at the costs tested; for the guess-test learner, lingering was 16–28% faster once switching cost anything, and with costs of two or three episodes per switch, active learning became slower than passive learning. The value of control therefore depends on what it costs to exercise. Like foraging animals deciding when to leave a patch (Charnov, 1976), a learner should stay with a scene while it still yields information at a higher rate than a new scene would, after the cost of finding one.

Limitations and future directions

Our two learners differ in instructive ways, but both are idealized, and Studies 2 and 3 use a single vocabulary size and Zipf exponent. Our mechanism tests for the guess-test learner rest on a few instrumented replications per condition, and we have not checked whether uniform targeting also helps the eliminative learner. Our policies remain idealized: the active learner knows exactly which words it has learned, the ramp in Study 3 is linear and indexed by episodes rather than by age, and the costs in Study 4 are free parameters. Flickr8k captions are adult descriptions of photographs, not child-directed speech or a child's view. Each of these idealizations could be informed by data. Egocentric recordings of infants' everyday experience (Clerkin et al., 2017; Sullivan et al., 2021) could be used to estimate how often children's attention is self-directed, which objects are available to choose among, how quickly self-directed exploration increases with motor development (Kretch et al., 2014), and how often a caregiver names a second object in the same scene rather than a new scene being sought. With such estimates, the present framework could predict not just whether active control helps, but how much each lever contributes over the course of a particular child's vocabulary growth.

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Table 1

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1000$-word vocabulary, eliminative learner, 30 replications per cell. Speedup is relative to Passive/Random at the same C. At $C = 100$, no random-context run reached even 50% of the vocabulary within the 5×10^8-episode cap, so speedups there are lower bounds.

	$C = 10$ ($C/W = 0.01$)		$C = 100$ ($C/W = 0.1$)	
Condition	Episodes	Speedup	Episodes	Speedup
Passive / Random	89,454	1x	$> 5 \times 10^8$	1x
Active / Random	4,695	19.1x	$> 5 \times 10^8$	–
Passive / Familiar	44,607	2.0x	446,962	>1,100x
Active / Familiar	3,127	28.6x	15,819	>31,000x

Table 2

Mean episodes to learn 99% of a W-word vocabulary at $C = 100$, eliminative learner. ">": no run reached even 10% (at $W = 1{,}000$, even 50%) of the vocabulary within the stated cap.

Condition	$W = 1{,}000$	$W = 10{,}000$	$W = 60{,}000$
Zipfian ($a = 1$), Passive / Random	$> 5 \times 10^8$	$> 10^8$	$> 10^8$
Zipfian ($a = 1$), Active / Random	$> 5 \times 10^8$	$> 10^8$	$> 10^8$
Zipfian ($a = 1$), Passive / Familiar	446,962	866,781	5,453,160
Zipfian ($a = 1$), Active / Familiar	15,819	113,584	589,504
Uniform, Passive / Random	9,904	80,048	424,968
Uniform, Active / Random	3,718	26,261	128,363

Table 3

Mean episodes to learn 99% of the vocabulary, eliminative learner, under frequency-weighted and uniform active target selection, and the corresponding speedups over passive selection in the same set of runs.

	Active (episodes)			Speedup over passive	
Setting	Freq.-weighted	Uniform	Passive	Freq.-weighted	Uniform
$W = 1{,}000$, $C = 10$, $a = 1$, random	4,657	4,670	88,220	18.9x	18.9x
$W = 10{,}000$, $C = 10$, $a = 1$, random	34,499	34,487	836,037	24.2x	24.2x
$W = 1{,}000$, $C = 100$, $a = 0.75$, random	632,415	631,345	6,843,752	10.8x	10.8x
$W = 1{,}000$, $C = 10$, $a = 1$, familiar	3,123	2,893	45,355	14.5x	15.7x
$W = 1{,}000$, $C = 100$, $a = 1$, familiar	15,978	8,181	474,823	29.7x	58.0x

Table 4

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1{,}000$-word vocabulary, eliminative learner, random context, with vs. without cross-word mutual exclusivity. Without mutual exclusivity, no $C = 100$ run reached even 50% of the vocabulary within 5×10^8 episodes, so those speedups are lower bounds.

Condition	Independent	Mutual exclusivity	Speedup
$C = 10$, Passive	89,454	23,184	3.9x
$C = 10$, Active	4,695	1,569	3.0x
$C = 100$, Passive	$> 5 \times 10^8$	771,125	>640x
$C = 100$, Active	$> 5 \times 10^8$	686,349	>720x

Table 5

Episodes to learn 99% of a $W = 1{,}000$-word Zipfian vocabulary, eliminative learner, relative to fully active selection in the same setting (first row: fully active episodes). Values above 1 are slower than fully active.

	$C = 10$, $a = 1$		$C = 100$, $a = 0.75$	$C = 100$, $a = 1$
Condition	Random	Familiar	Random	Familiar
Fully active (episodes)	4,692	3,132	628,691	16,060
Passive	19.0x	14.2x	10.6x	26.6x
25% active	2.67x	2.51x	2.33x	1.81x
50% active	1.67x	1.63x	1.58x	1.34x
75% active	1.24x	1.23x	1.21x	1.11x
Goldilocks	1.00x	1.04x	1.00x	0.56x
Choice set of 100	1.03x	1.09x	1.03x	5.51x
Choice set of 50	1.13x	1.15x	1.09x	8.01x
Choice set of 20	1.54x	1.43x	1.32x	11.3x
Choice set of 5	4.15x	3.32x	2.78x	10.4x

Table 6

Episodes to learn 99% of a $W = 1{,}000$-word Zipfian ($a = 1$) vocabulary at $C = 100$ with familiar context, relative to unrestricted active selection under the same familiarity measure (30 replications per cell). "All words": the original familiarity measure, in which every unencountered word counts as still holding every referent; "encountered words only": that contribution removed.

	Familiarity computed from	
Condition	All words	Encountered words only
Unrestricted active (episodes)	15,770	7,803
Choice set of 100, frequency-weighted	5.66x	1.34x
Choice set of 20, frequency-weighted	11.2x	3.32x
Choice set of 100, uniform	0.96x	1.01x
Goldilocks	0.57x	0.93x
Passive	29.6x	57.9x

Table 7

Mean episodes to learn 99% of a $W = 1{,}000$-word Zipfian ($a = 1$) vocabulary, guess-test learner, 30 replications per cell. ">" marks cells in which some runs did not finish within 5×10^6 episodes (number in parentheses); these runs are counted at the cap, so the mean is a lower bound.

	$C = 10$		$C = 100$	
Condition	Random	Familiar	Random	Familiar
Passive	39,327	26,382	328,648	85,818
25% active	15,218	8,218	255,740	68,658
50% active	11,460	6,387	237,110	71,732
75% active	9,270	5,940	200,699	80,128
Fully active	7,814	7,787	179,983	>3,718,842 (22)
Goldilocks	3,632	3,594	63,772	>4,835,958 (29)
Choice set of 100	10,333	17,322	1,589,313	>2,235,324 (7)
Choice set of 50	13,360	26,275	2,793,079	>3,582,840 (7)
Choice set of 20	6,134	15,897	>2,668,049 (4)	>3,570,502 (15)
Choice set of 5	10,979	9,102	501,200	350,426

Table 8

Mean episodes to learn 99% of a Zipfian ($a = 1$), $W = 1{,}000$-word vocabulary, eliminative learner, $C = 10$, for constant levels of autonomy (proportion of episodes on which the target is actively selected); 30 replications each. Study 2 reports some of the same blends, relative to fully active selection, from a separate set of simulations.

Proportion active	Episodes	Proportion active	Episodes
0 (passive)	89,857	0.30	11,048
0.02	41,853	0.40	9,190
0.05	29,433	0.50	7,803
0.075	24,317	0.60	6,869
0.10	21,116	0.75	5,829
0.15	16,871	0.90	5,073
0.20	14,332	1 (active)	4,681
0.25	12,633		

Table 9

Mean episodes to reach 50/90/99% of the 3,598-word learnable vocabulary, Flickr8k captions, eliminative mechanism. `move_cost` *is the number of extra episodes charged for switching to a new photo (passive sampling has no such switching decision).*

Condition	move_cost	50%	90%	99%
Passive	–	10,386	60,719	151,671
Active, move-on	0	2,779	5,643	6,528
Active, linger	0	3,125	6,511	7,515
Active, move-on	1	5,558	11,285	13,057
Active, linger	1	5,483	11,587	13,461
Active, move-on	2	8,338	16,928	19,585
Active, linger	2	7,841	16,664	19,408
Active, move-on	3	11,117	22,570	26,113
Active, linger	3	10,199	21,740	25,354
Active, move-on	5	16,675	33,856	39,170
Active, linger	5	14,915	31,893	37,247
Active, move-on	10	30,571	62,069	71,811
Active, linger	10	26,705	57,275	66,980

Table 10

Guess-test learner on Flickr8k captions: mean episodes to learn 50%, 90%, and 95% of the 3,598-word learnable vocabulary, and the proportion learned at the end of the run (cap 10^6 *episodes), 30 replications per cell. Passive learning reached 99% in every run (mean 165,506 episodes); no active run did.*

Condition	move_cost	50%	90%	95%	Final
Passive	–	7,746	39,349	60,015	99.3%
Active, move-on	0	6,291	17,165	21,937	97.9%
Active, linger	0	6,693	17,976	21,862	98.5%
Active, move-on	1	12,583	34,330	43,875	97.9%
Active, linger	1	9,631	29,663	37,030	98.5%
Active, move-on	2	18,874	51,496	65,812	97.9%
Active, linger	2	12,569	41,351	52,199	98.5%
Active, move-on	3	25,166	68,661	87,749	97.9%
Active, linger	3	15,507	53,039	67,367	98.5%
Active, move-on	5	37,749	102,991	131,623	97.9%
Active, linger	5	21,383	76,414	97,704	98.5%
Active, move-on	10	69,206	188,817	241,310	97.9%
Active, linger	10	36,073	134,853	173,547	98.5%

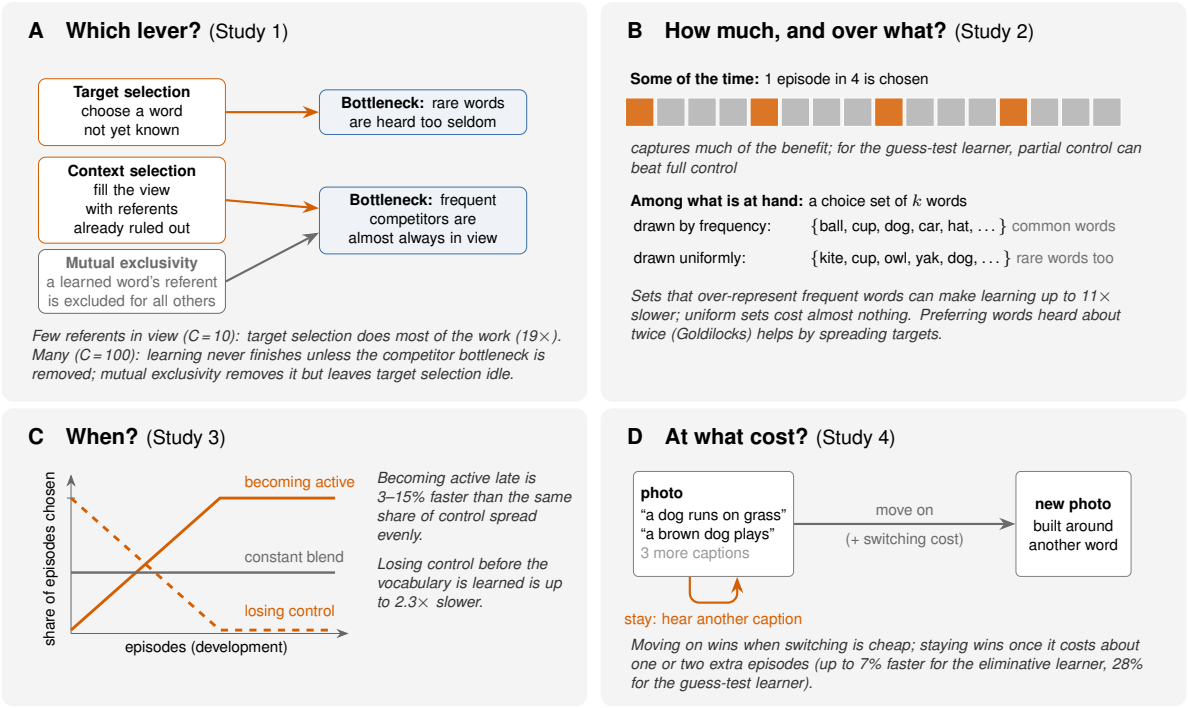


Figure 1. Overview of the four studies. (A) An eliminative learner can be held up by rare words that are heard too seldom, or by frequent competitors that are almost always in view; target selection acts on the first, and context selection or cross-word mutual exclusivity on the second. (B) A learner may exercise control only on some episodes, and only over a set of words at hand. (C) Control may arrive or fade over development. (D) On real captions, a learner can stay with the current photo for another caption or move to a new one at a cost. Orange marks active choices, grey passive input or properties of the learner; italic notes give the main result for each study.

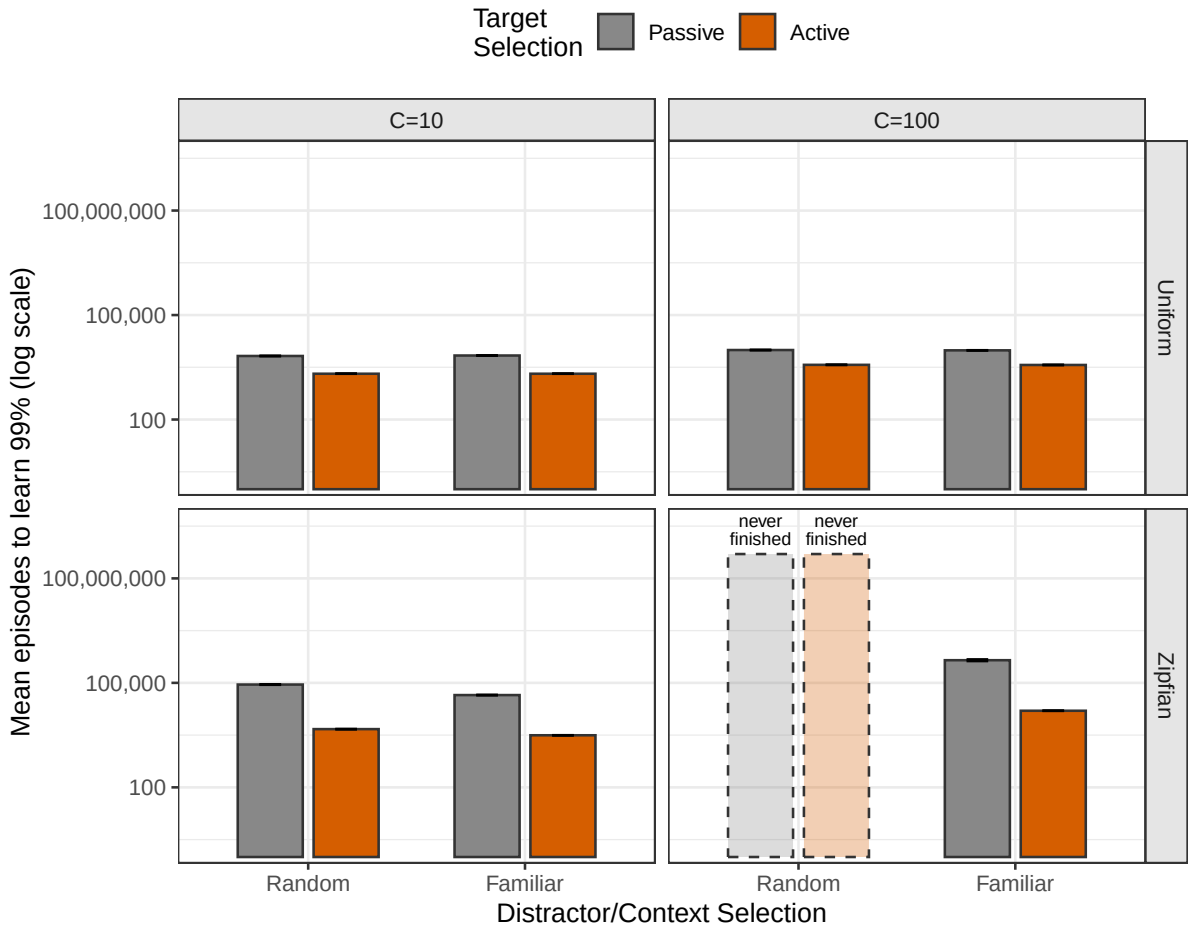


Figure 2. Mean episodes (log scale, ± 1 SE) for the eliminative learner to acquire 99% of a 1000-word vocabulary, by target selection (bar fill) and context selection (x-axis), for uniform and Zipfian ($a = 1$) distributions and $C = 10$ and $C = 100$. Faded, dashed bars mark conditions in which no run reached 99% (or even 50%) of the vocabulary within the 5×10^8-episode cap; they are drawn at the cap and are lower bounds.

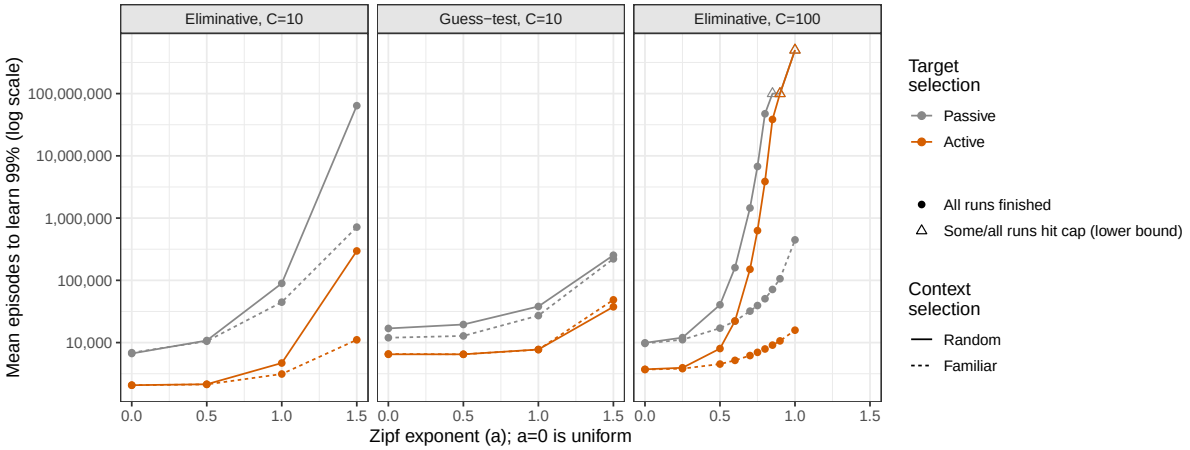


Figure 3. Mean episodes (log scale) to acquire 99% of a 1000-word vocabulary as a function of the Zipf exponent a, for the eliminative and guess-test learners at $C = 10$ and the eliminative learner at $C = 100$, by target selection (color) and context selection (line type). Open triangles mark cells in which some or all runs reached the episode cap (5×10^7 for $a \leq 0.75$, 10^8 for $0.8 \leq a \leq 0.9$, 5×10^8 for $a = 1$); these points are lower bounds.

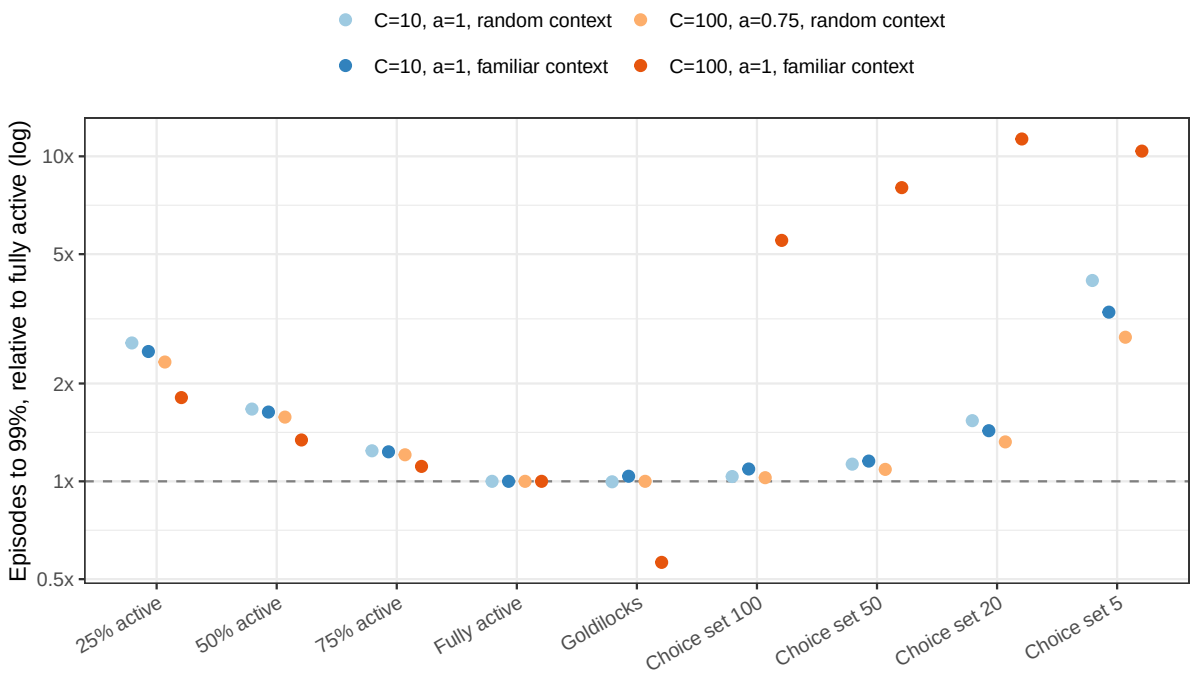


Figure 4. Episodes to learn 99% of the vocabulary relative to fully active selection (log scale; dashed line = fully active), eliminative learner, for partial autonomy, Goldilocks targeting, and bounded choice sets, in four settings. Pure passive sampling (10.6–26.6x) is omitted for scale.

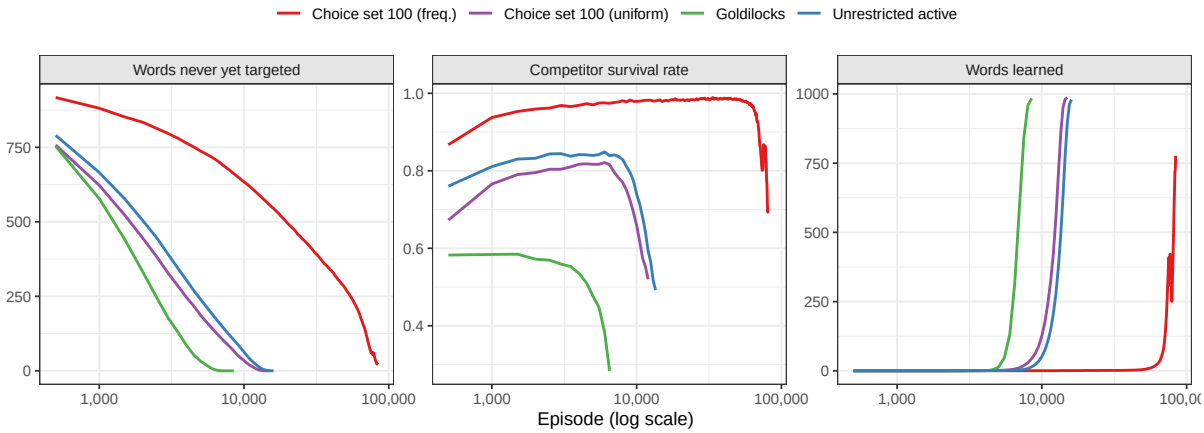


Figure 5. Trajectories under the original familiarity measure, $C = 100$, $a = 1$, familiar context (means of 10 runs). Left: words never yet targeted. Middle: competitor survival, the share of the target's remaining competing referents that reappear among its distractors and so escape elimination, shown until half the vocabulary is learned. Right: words learned. The frequency-weighted choice set leaves many words unencountered for longer, and while it does, familiar-context selection barely reduces competitor survival.

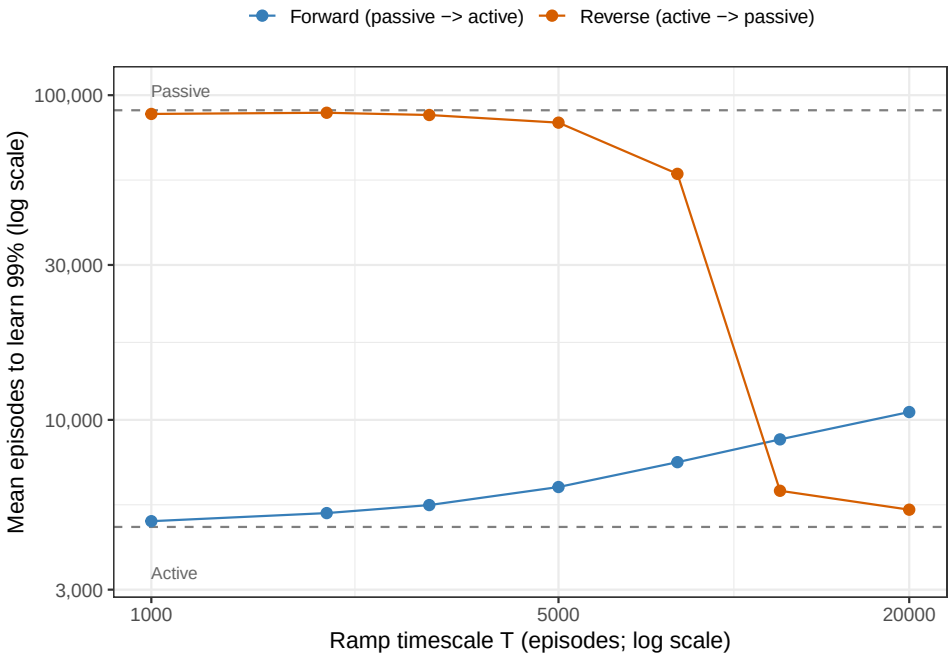


Figure 6. Mean episodes to learn 99% of the vocabulary vs. ramp timescale T, forward (passive-to-active) vs. reverse (active-to-passive) developmental trajectories. Dashed lines mark pure passive and pure active selection (Table 8).

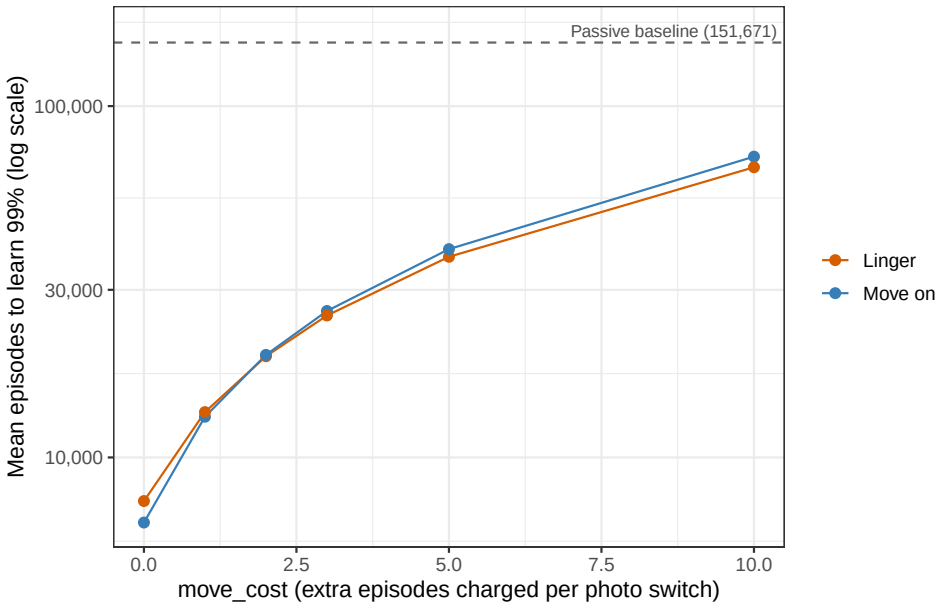


Figure 7. Mean episodes to learn 99% of the vocabulary vs. `move_cost`, active eliminative learner on Flickr8k captions. Moving on to a new photo wins when switching is cheap; lingering on the current photo overtakes it once switching costs more, with the crossover between `move_cost`=1 and 2.